

Department of Physics-National Institute of Technology, Tiruchirappalli

Date: 26/11/2025 & Time: 3.00 – 4.00 pm CT-II

Subject: Physics-I (PHIR11)

Branch and Section: ICE-B

Answer all Questions

20 Marks

1. (a) Describe the major limitation of the Rayleigh–Jeans law in explaining blackbody radiation and illustrate your answer with a suitable diagram (b) Explain the four laws associated with blackbody radiation. (3)
2. If the kinetic energy of an electron known to be about 1 eV, must be measured to within 0.0001 eV, what accuracy can its position be measured simultaneously? (2)
3. Discuss Einstein's explanation of the photoelectric effect in detail, supported with a neat diagram. (3)
4. A particle is restricted to move in one dimension between two infinitely high potential barriers at $x=0$ and $x=L$, where the potential is zero inside. Use the Schrödinger time-independent equation to determine the quantized energy levels, probability density and the normalized wave functions of the particle with neat diagram. (5)
5. Explain salient features of nuclear shell model. What are magic numbers given for nucleus of an atom? (2)

Write down all correct option(s) (Each carries 1 mark)

(5)

6. Which of the following is NOT a property of nuclear forces?
(a) They are attractive at long distances and repulsive at short distances
(b) They are charge-dependent
(c) They obey saturation
(d) They are spin-dependent
7. Which nucleus has a non-zero magnetic moment?
(a) ^{12}C
(b) ^{16}O
(c) ^{14}N
(d) ^4He
8. Which of the following is not the characteristics of Planck's black body radiation distribution? (Multiple Ans correct)
(a) As temperature increases the peak of the curve shift towards higher wavelength
(b) Spectral emissive power varies continuously with the change in the wavelength
(c) At a given wave length, emissive power increases with temperature
(d) As temperature increases the peak of the curve shift towards lower wavelength
9. A radioactive nucleus decays by 40% in 10 minutes. What is the decay constant?
(a) 0.051 min^{-1}
(b) 0.040 min^{-1}
(c) 0.010 min^{-1}
(d) 0.023 min^{-1}
10. A particle has a position uncertainty of $\Delta x = 1 \times 10^{-10} \text{ m}$. What is the minimum uncertainty in momentum?
(a) $1.0 \times 10^{-24} \text{ kgm/s}$
(b) $5.3 \times 10^{-25} \text{ kgm/s}$
(c) $7.0 \times 10^{-26} \text{ kgm/s}$
(d) $1.0 \times 10^{-21} \text{ kgm/s}$

[Constants: $\hbar = 1.05 \times 10^{-34} \text{ Js}$; $e = 1.602 \times 10^{-19} \text{ C}$; $h = 6.626 \times 10^{-34} \text{ Js}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$]
